OS & SP

Name : S LOHITHA **Roll No :** 2520030109

SKILL – 14

1.Skill14.c

lohitha27@LAPTOP-A9F545F(

GNU nano 8.7.1 skill14.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
    int output_fd;
    int error_fd;
    pid_t pid;

    /* Open append.txt in append mode */
    output_fd = open(
        "append.txt",
        O_WRONLY | O_CREAT | O_APPEND,
        0644
    );

    if (output_fd == -1) {
        perror("Error opening append.txt");
        return 1;
    }

    /* Open errors.txt */
    error_fd = open(
        "errors.txt",
        O_WRONLY | O_CREAT | O_TRUNC,
        0644
    );

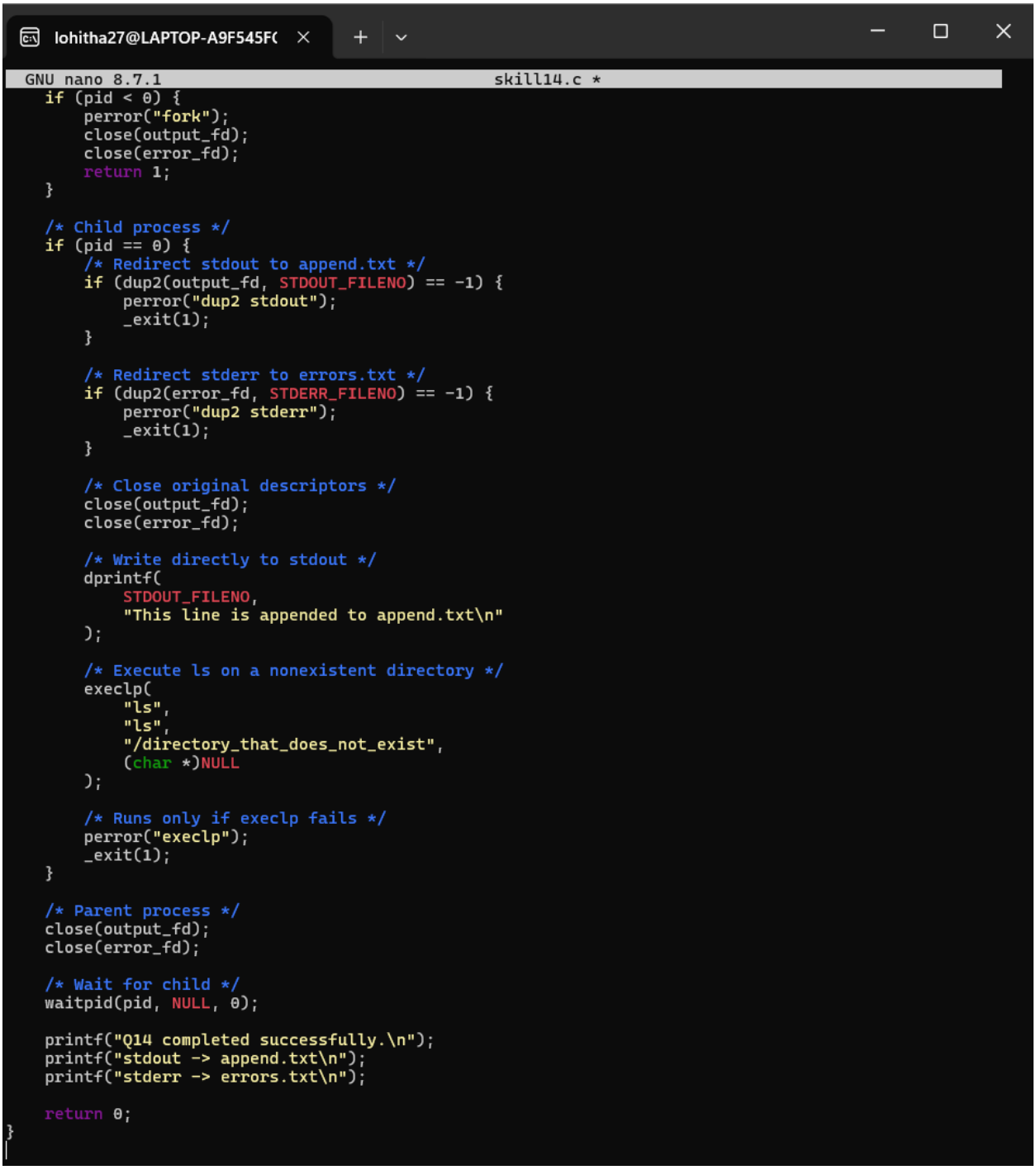
    if (error_fd == -1) {
        perror("Error opening errors.txt");
        close(output_fd);
        return 1;
    }

    /* Create child process */
    pid = fork();

    if (pid < 0) {
        perror("fork");
        close(output_fd);
        close(error_fd);
        return 1;
    }

    /* Child process */
    if (pid == 0) {
        /* Redirect stdout to append.txt */
        if (dup2(output_fd, STDOUT_FILENO) == -1) {
            perror("dup2 stdout");
            _exit(1);
        }

        /* Redirect stderr to errors.txt */
        if (dup2(error_fd, STDERR_FILENO) == -1) {
            perror("dup2 stderr");
```


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GNU nano 8.7.1 skill14.c *
if (pid < 0) {
perror("fork");
close(output_fd);
close(error_fd);
return 1;
}
/* Child process */
if (pid == 0) {
/* Redirect stdout to append.txt */
if (dup2(output_fd, STDOUT_FILENO) == -1) {
perror("dup2 stdout");
_exit(1);
}
/* Redirect stderr to errors.txt */
if (dup2(error_fd, STDERR_FILENO) == -1) {
perror("dup2 stderr");
_exit(1);
}
/* Close original descriptors */
close(output_fd);
close(error_fd);
/* Write directly to stdout */
dprintf(
STDOUT_FILENO,
"This line is appended to append.txt\n"
);
/* Execute ls on a nonexistent directory */
execlp(
"ls",
"ls",
"/directory_that_does_not_exist",
(char *)NULL
);
/* Runs only if execlp fails */
perror("execlp");
_exit(1);
}
/* Parent process */
close(output_fd);
close(error_fd);
/* Wait for child */
waitpid(pid, NULL, 0);
printf("Q14 completed successfully.\n");
printf("stdout -> append.txt\n");
printf("stderr -> errors.txt\n");
return 0;
}

OUTPUT :

```
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill14.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc -Wall -Wextra -g skill14.c -o skill14
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill14
Q14 completed successfully.
stdout -> append.txt
stderr -> errors.txt
lohitha27@LAPTOP-A9F545FG:~/SKILL$
```